

Motor Failures Prediction.

*What if motor failure could be detected before the
breakdown occurs?*

K-Nearest Neighbors · Condition Monitoring · Predictive Maintenance



>1,800

Motor Records

6

Sensor Signals

88.3%

Accuracy

0.940

ROC-AUC

A motor does not fail suddenly — it degrades through measurable physical signals for weeks before shutdown.



Cost

Unplanned downtime, emergency repairs, lost production hours



Safety

Motor failures in critical systems create hazardous conditions



Operations

Cascading failures, maintenance bottlenecks, line disruptions

29.0%

of motor records show incipient failure signals — detectable in sensor data long before catastrophic breakdown.

Old Thinking

Failures are sudden events:

"It tripped / it didn't."

Reactive maintenance.
Fix after breakdown.
No detection. No warning.
Just unplanned downtime.



New Thinking

Failure is a probabilistic state:

"How likely is incipient failure now?"

Score each motor continuously.
Flag degradation before shutdown.
Schedule maintenance proactively.
Convert signals into decisions.

Six condition-monitoring signals — each one capturing a different physical dimension of motor degradation:

VIB RMS	Vibration RMS (mm/s) <i>Overall vibration energy — primary indicator of mechanical degradation. Normal: ~2.2 Failure: ~4.3</i>
VIB P-P	Vibration Peak-to-Peak (mm/s) <i>Captures impact events: looseness, imbalance, bearing defects. Normal: ~5.0 Failure: ~9.0</i>
BEARING T	Bearing Temperature (°C) <i>Friction buildup from lubrication loss or overloading. Normal: ~60°C Failure: ~82°C</i>
CURRENT	Motor Current (A) <i>Overcurrent indicates increased mechanical resistance. Normal: ~28A Failure: ~35A</i>
FREQ	Dominant Frequency (Hz) <i>Frequency drop signals misalignment or mechanical looseness. Normal: ~29.5Hz Failure: ~24.5Hz</i>
LOAD %	Load Percentage (%) <i>Operating load context — high load amplifies all other degradation signals. Range: 41–111%</i>

KNN is ideal here — motor physics says that similar sensor profiles should share similar operating states.

If two motors have similar vibration, temperature, and current profiles — they're likely in the same operating state.

Distance = Physics

KNN finds the k most similar historical motor states. Physical similarity maps directly to operational similarity — an intuition engineers share.

Non-linear Boundaries

Motor degradation creates irregular decision boundaries in sensor space. KNN adapts naturally — no linear assumption required.

Scaling is Essential

StandardScaler prevents Ampere-scale current from dominating mm/s vibration. Preprocessing inside Pipeline prevents data leakage.

GridSearchCV Tuning

Optimal $k=7$, $\text{weights}='distance'$, $p=1$ (Manhattan). Distance weighting gives closer neighbors more diagnostic influence.

88.3%

Accuracy

0.940

ROC-AUC

71.4%

Recall (Failure)

0.780

F1 (Failure)

Permutation Importance (accuracy drop when shuffled)

vib_rms_mms	$\Delta -1.3\%$
vib_peak_to_peak_mms	$\Delta -0.7\%$
bearing_temp_c	$\Delta +0.1\%$
motor_current_a	$\Delta -0.1\%$
dominant_freq_hz	$\Delta -0.0\%$
load_pct	$\Delta -0.1\%$

Confusion Matrix (555 test records)

	Pred: Normal	Pred: Failure
Actual: Normal	375 (TN)	19 (FP)
Actual: Failure	46 (FN)	115 (TP)

46 failures missed. In predictive maintenance, each FN represents an undetected degradation risk.

AUC 0.940 — the model nearly perfectly ranks failing motors above healthy ones across all thresholds.

VIB RMS Δ -1.3%

→ Primary alarm trigger. Set continuous monitoring threshold at 3.0 mm/s — anything above initiates inspection scheduling.

VIB P-P Δ -0.7%

→ Impact event detector. Rising peak-to-peak alongside stable RMS signals looseness or early bearing defects specifically.

Bearing T Δ +0.1%

→ Thermal confirmation signal. Above 72°C with elevated vibration = strong failure indicator. Use as secondary threshold.

Motor Current Δ -0.1%

→ Overcurrent amplifies mechanical signals. Combine with vibration to distinguish electrical from mechanical degradation.

Freq Δ -0.0%

→ Contextual signal — frequency drop confirms misalignment type. Low individual importance but useful for failure classification.

Same model. Three motors. One shows the boundary — where the model switches from Normal to Failure.

Motor A — Healthy	Motor B — Failure	Motor C — Borderline
Vib RMS2.0 mm/s	Vib RMS4.8 mm/s	Vib RMS3.2 mm/s
Vib P-P5.0 mm/s	Vib P-P9.8 mm/s	Vib P-P6.5 mm/s
Bearing60°C	Bearing88°C	Bearing68°C
Current28A	Current37A	Current30A
Freq30 Hz	Freq23 Hz	Freq28.5 Hz
Load70%	Load92%	Load78%
2% failure probability	98% failure probability	45% failure probability

Motor C insight: elevated vibration at 3.2 mm/s is sufficient to place this motor at 45% — continuous monitoring is warranted.

LOZANOLSA · PROJECT 83

Motor Failure Predictor

KNN trained on 1,500 motor condition records · 6 sensors

// VIBRATION

RMS (MM/S)

2.20

PEAK-TO-PEAK (MM/S)

12.30

// THERMAL & ELECTRICAL

BEARING TEMP (°C)

53.50

MOTOR CURRENT (A)

35.50

// SPECTRAL & LOAD

DOMINANT FREQ (HZ)

ML PROJECT #03 · K-NEAREST NEIGHBORS · PREDICTIVE MAINTENANCE

Motor Failures Are Not Random

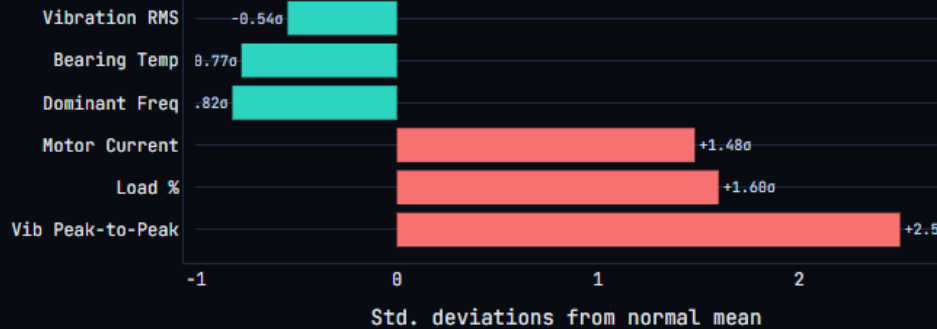
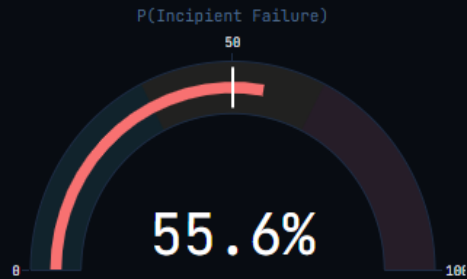
Distance in sensor space is operational similarity.

KNN 6 SENSORS 97.8% ACCURACY AUC 0.996 GRIDSEARCHCV

DATA EXPLORER PERFORMANCE SIMULATOR RISK DRIVERS ACTION PLAN

// FAILURE PROBABILITY

// DEVIATION FROM NORMAL-CLASS PROFILE



INCIPIENT FAILURE — SCHEDULE INSPECTION

Priority: MEDIUM · Increase monitoring — next 48h

This project is not about predicting motor failures.

It is about catching them while there's still time.

KNN doesn't predict the future —
it finds the motors whose present looks like past failures.

Feel free to download the full project: dataset, notebook, and simulator.

Change k . Try different distance metrics. Adjust the thresholds.

Because machine reliability has no finish line.

